Solving Matrix Games

Book: Algorithmic Game Theory

Lemke-Howson Alg.

Finding NE (Review)

P2

P1	stag	hare
stag	10,10	0,6
hare	6,0	5,5

Nash Equilibrium

Every player plays a best response

$M \equiv$ set of actions for P1 $\qquad N \equiv$ set of actions for P2

Bimatrix Games

$$A = \begin{bmatrix} 10 & 0 \\ 6 & 5 \end{bmatrix} \qquad B = \begin{bmatrix} 10 \leftarrow & 6 \\ 0 \rightarrow & 5 \end{bmatrix}$$

$x \equiv$ strategy for P1
$y \equiv$ strategy for P2

payoffs: P1: $x^T A y$
P2: $x^T B y$

Calculate Mixed NE

To get another player to play a mixed NE, you must make them indifferent between their actions.

P1 indifferent

$$\begin{array}{l} \text{payoff for P1 playing 1} \rightarrow \\ \text{payoff for P1 playing 2} \rightarrow \end{array} \begin{bmatrix} u \\ u \end{bmatrix} = \overset{A}{\begin{bmatrix} 10 & 0 \\ 6 & 5 \end{bmatrix}} \overset{y}{\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}}$$

$y_1 + y_2 = 1$

$\boxed{y_1 = \frac{5}{9} \quad y_2 = \frac{4}{9}}$

P2 indifferent

$$\overset{x^T}{\begin{bmatrix} x_1 & x_2 \end{bmatrix}} \overset{B}{\begin{bmatrix} 10 & 6 \\ 0 & 5 \end{bmatrix}} = \begin{bmatrix} v & v \end{bmatrix}$$

$x_1 + x_2 = 1$

$\boxed{x_1 = \frac{5}{9} \quad x_2 = \frac{4}{9}}$

$M = \{1, 2, 3\}$ $\qquad N = \{4, 5\}$

$$A = \begin{array}{c} 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} \boxed{3} & 3 \\ 2 & 5 \\ 0 & 6 \end{bmatrix} \qquad B = \begin{array}{c} \\ \\ \\ \end{array} \begin{array}{cc} 4 & 5 \\ \end{array}$$

$$B = \begin{array}{cc} 4 & 5 \\ \hline \end{array} \begin{bmatrix} \boxed{3} \leftarrow & 2 \\ 2 \rightarrow & 6 \\ 3 \leftarrow & 1 \end{bmatrix}$$

Pure NE: $x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $\quad y = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Mixed NE

P1 indifferent

$$\begin{bmatrix} u \\ u \\ u \end{bmatrix} = Ay$$

$y_4 + y_5 = 1$

4 equations 3 unknowns

In general, could have solutions, but for this A, no solutions.

No mixed strategies where P1 has all actions in the support of the NE

What if 1 of P1's actions is not in the support of the NE.

[Every action in the support of a NE must be a best response

Mathematically, in a NE:

$$x_i > 0 \Rightarrow (Ay)_i = u = \max\{(Ay)_k : k \in M\}$$
$$y_j > 0 \Rightarrow (x^T B)_j = v = \max\{(x^T B)_k : k \in N\}$$

$M = \{1, 2, 3\}$ $\qquad\qquad$ $N = \{4, 5\}$

$$A = \begin{array}{c} 1 \\ 2 \\ 3 \end{array} \begin{bmatrix} \boxed{3} & 3 \\ 2 & 5 \\ 0 & 6 \end{bmatrix} \qquad B = \begin{array}{cc} 4 & 5 \\ \left[\begin{matrix} \boxed{3} & 2 \\ 2 & 6 \\ 3 & 1 \end{matrix}\right] \end{array}$$

Pure NE: $x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $\quad y = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Mixed NE: $x = [4/5, \frac{1}{5}, 0]^T$ $\quad y = [2/3, \frac{1}{3}]^T$

$Ay = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix}$ $\quad x^T B = [14/5, \frac{14}{5}]$

$x = [0, \frac{1}{3}, \frac{2}{3}]^T$ $\quad y = [1/3, \frac{2}{3}]^T$

$Ay = \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix}$ $\quad x^T B = [8/3, 8/3]$

Hard part of finding NE in general Matrix Games: Finding Support

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First, narrow scope to avoid cases with possibly infinite NE

Def. A two-player game is called nondegenerate if no mixed strategy of support size k has more than k pure best responses.

Example of degenerate

$$A = \begin{bmatrix} 3 & 3 \\ 2 & 5 \\ 0 & 6 \end{bmatrix} \qquad B = \begin{bmatrix} 3 & 2 \\ 2 & 6 \\ 3 & 1 \end{bmatrix}$$